

Mohammed Ismail Sarfaraz Shaik

+1 (202) 684-0434 | m.shaik@gwu.edu | LinkedIn | GitHub | Portfolio

SUMMARY

Data Analyst with 2+ years of end-to-end experience analyzing large datasets to surface insights driving business outcomes across healthcare and fintech. Skilled in SQL, Python, Tableau & statistical analysis (ANOVA, A/B testing) to build clean datasets, develop executive dashboards & enable decision-making at scale. Proven ability to translate complex data into clear insights for stakeholders. Authorized to work in the USA.

TECHNICAL SKILLS

Languages: SQL, Python (Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn), R, NoSQL, SAS

BI & Data Tools: Tableau, PowerBI, Excel, Looker, dbt, Databricks, Snowflake, BigQuery, KPI Development, PySpark, ETL/ELT

Analytics & Statistics: A/B Testing, Hypothesis Testing, Regression Analysis, ANOVA, Cohort Analysis, Funnel Analysis, Statistical Modeling

Cloud & ML: AWS (S3, Athena, Redshift, QuickSight, Glue), Predictive Analytics, Airflow, Docker, Hugging Face, LangChain, RAG, Git

PROFESSIONAL EXPERIENCE

George Washington University

Data Analyst

Mar 2025 - Present

Washington, DC, USA

- Reconciled \$2M+ in historical funding allocations, fixing corrupted inputs to departmental forecasting models for accurate projections.
- Authored analytical SQL queries & Python validation pipelines on multi-source grant & cohort data, powering live executive dashboards.
- Applied ANOVA on demographic & program variables to identify retention drivers, informing redesign that closed 25% attrition gap.

Tata Consultancy Services (TCS)

Software Engineer (Data & ML)

Jun 2023 - Jun 2024

Hyderabad, TG, India

- Built curated feature tables (PySpark) supporting patient readmission and dialysis risk analytics for chronic care reporting programs.
- Tracked downstream model AUC lift from 0.76 to 0.84 by improving feature quality & eliminating upstream gaps in data ingestion.
- Validated new pipeline via 30-day A/B test, tracking feature drift, model lift & SLA compliance against legacy heuristics before cutover.
- Architected Medallion ETL pipeline (Databricks/Airflow) syncing 12 PHI feeds to Snowflake for real-time clinical reporting & analytics.
- Designed HIPAA-compliant validation framework (Great Expectations) on feature tables, cutting quality incidents 20% in production.

Tata Consultancy Services (TCS)

Data Analyst Intern

Mar 2023 - May 2023

Hyderabad, TG, India

- Conducted Pearson correlation analysis across 20+ metrics to define 15+ strategic KPIs aligned with core business strategy and risk metrics.
- Delivered business intelligence solutions for a global capital markets client via Tableau dashboards with LOD expressions and parameters.
- Tuned SQL queries for OLTP systems (50M+ daily transactions), improving data retrieval & powering real-time performance dashboards.

PROJECTS

AWS CloudGuide - Cloud Architecture Assistant - [Link to project](#)

- Architected hybrid RAG using dense (bge-m3) & sparse (BM25) retrieval via RRF, resolving semantic ambiguity across 100k+ AWS pages.
- Scaled batched GPU pipeline (PyTorch, T4) handling 800k+ vectors using recursive chunking for context preservation & retrieval accuracy.
- Deployed LLM-powered conversational Q&A system (Qwen-2.5-7B) via Hugging Face achieving sub-200ms latency for cited answers.

Automatic Image Restoration & Colorization using Deep Convolutional Autoencoders - [Link to project](#)

- Orchestrated scalable PyTorch pipeline on AWS EC2 processing 118k COCO images using Mixed Precision to optimize training.
- Pioneered a ResNet-18 U-Net & VAE-regularized framework with Transfer Learning, achieving 85% model compression and 0.496 SSIM.

IPL Live Match Win Predictor - [Link to project](#)

- Developed a live, ball-by-ball win probability predictor for IPL matches using an XGBoost model trained on 500+ historical matches.
- Stabilized early-game prediction volatility by designing a confidence-weighted blend-in algorithm, reducing forecast uncertainty by 60%.

EDUCATION

George Washington University

Master's, Data Science

Aug 2024 - May 2026

GPA: 3.95

- Relevant Coursework: Data Mining, Deep Learning, Mathematics in Machine Learning, Data Analytics, Data Visualization, NLP